

iFogSim: An Open Source Simulator *for* Edge Computing, Fog Computing and IoT



Typically, within any organisation, numerous devices and gadgets are connected via advanced wireless technologies for different applications like communication, surveillance, security, messaging, social media, defence, etc. These deployments are traditionally associated with different paradigms including the Internet of Things (IoT), fog computing, edge computing and cloud computing. All these deployments, which need effective and high performance network environments and aware communications, have opened up areas of research.

IoT refers to the smart objects and gadgets that are connected using wireless technologies for real-time communication and transmission of signals. Different research reports predict that there will be more than 30 billion connected devices by 2020. In addition, the IoT market is expected to grow to over US\$ 8 trillion by that year.

Cloud computing refers to the delivery of computing services using Internet based delivery channels. Remote data centres that provide services via the Platform as a Service (PaaS), Infrastructure as a Service (IaaS), Software as a Service (SaaS) and other related service delivery models are maintained in the cloud. Such services depend a lot on Internet connectivity, and performance is related to the bandwidth and channels available. Cloud computing is

entirely dependent on Internet speed and connections, which may not be very good at every location.

Fog computing

Fog computing is gaining prominence in network based smart environments. It has been launched by Cisco with the aim of bringing cloud computing nearer to the control and management region of the end users. Fog computing enables users to access services and objects without depending on the Internet. It is closer to the users accessing the services, which makes the access speed very fast compared to cloud computing based delivery. Fog computing is also associated with the terms 'fogging' and fog networking.

In fog computing, a decentralised computing environment is established. In this approach, the services, objects, storage, applications and data analytics are deployed between the local systems (nodes or edges) and remote data centre. In this way, the actual resources being used are nearer to the user and there is no need to access everything via the Internet.

Fog = Cloud + Internet of Things (IoT)

Fog computing is the intermediate layer between the remote cloud and the end user, and addresses the issues of network bandwidth, latency, delay and jitter — all of which can